VICTORY 2027

PHOTOSYNTHESIS
(PHOTOSYNTHESIS PROCESS)

LECTURE- 04 BIOLOGY

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TOPICS TO BE COVERED

Sugar concentration theory

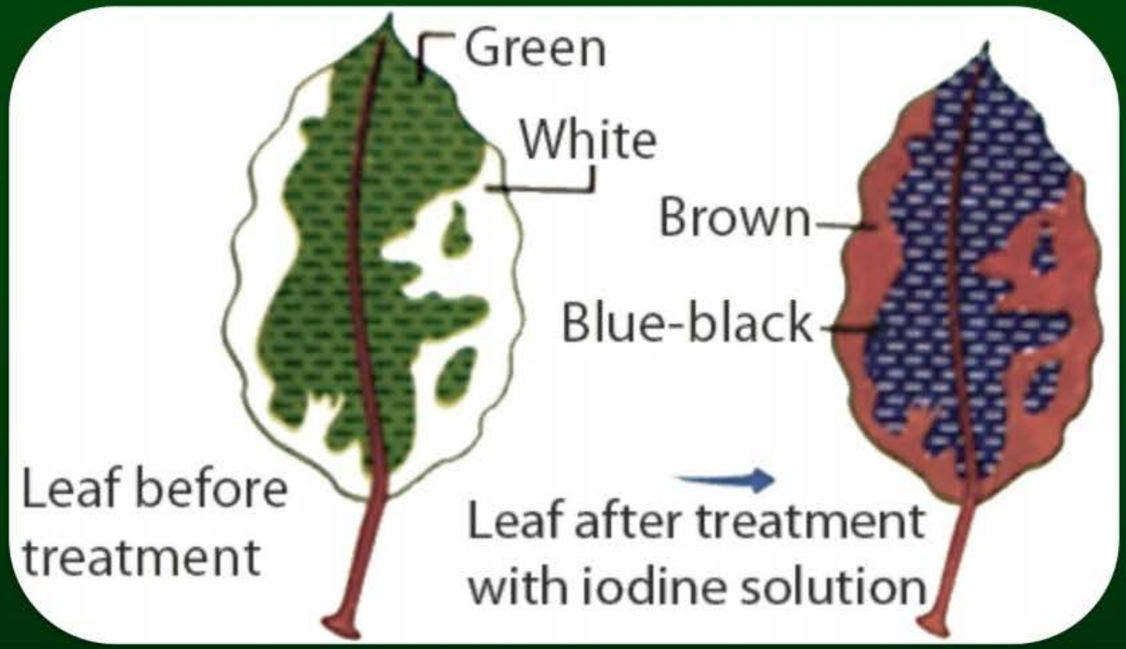
Experiments of Photosynthesis

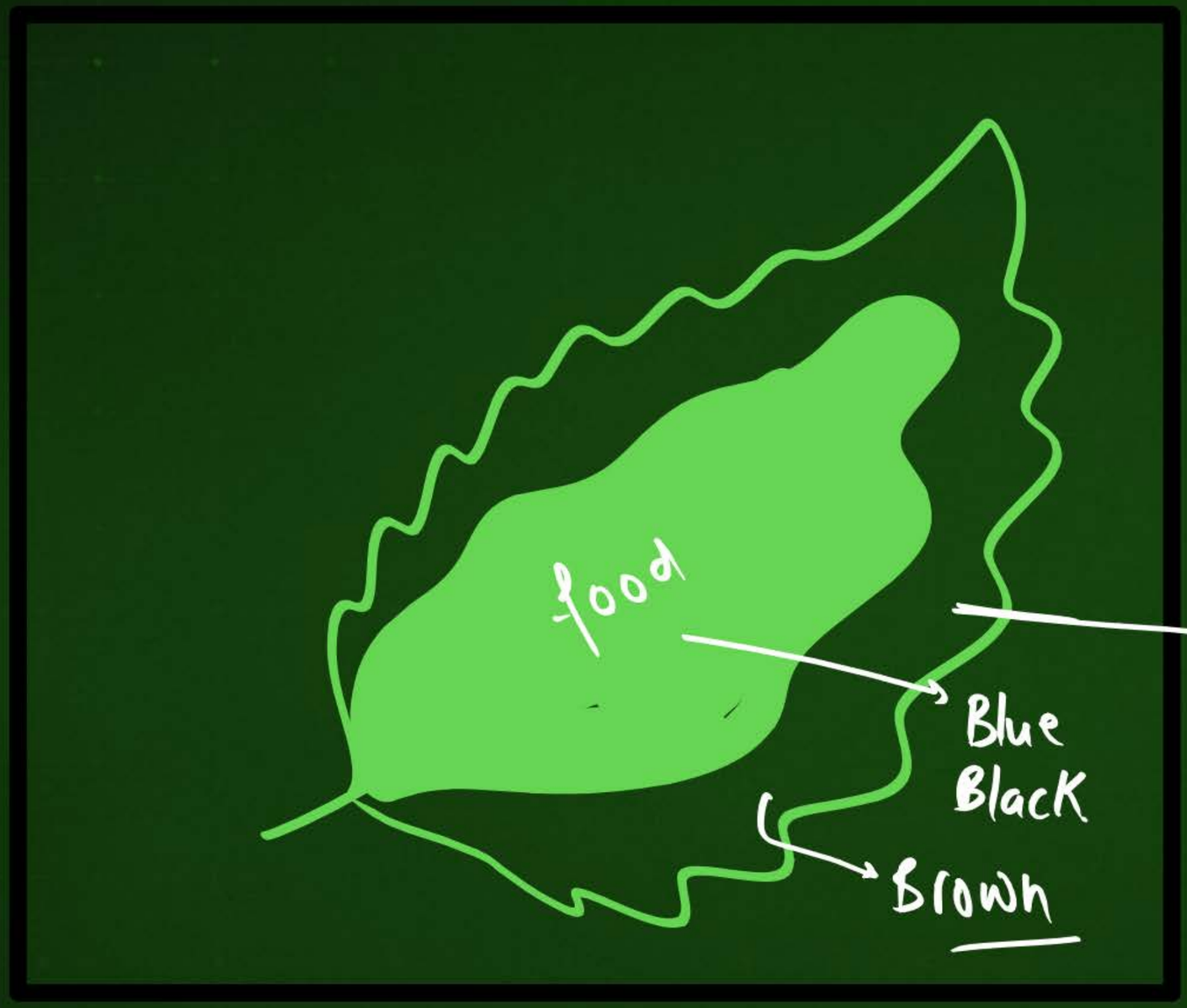

HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上



Experiment 1: To show that chlorophyll is Necessary for Photosynthesis

- Take a plant with variegated leaves having some green and some non-green areas. **Examples:** Coleus, Geranium and Croton.
- Destarch the leaves by keeping the plant in a dark room for a few days.
- Place the plant in the sun.
- After a few hours, pluck one leaf.
- Make its outline on paper and mark the green and non-green areas on the outline.
- Test the leaf for starch. Only the green parts of the leaf turn bluish, showing the presence of starch.





Destarch

All the starch will be used up.

Non green part

IODINE

↓

All BROWN

↓

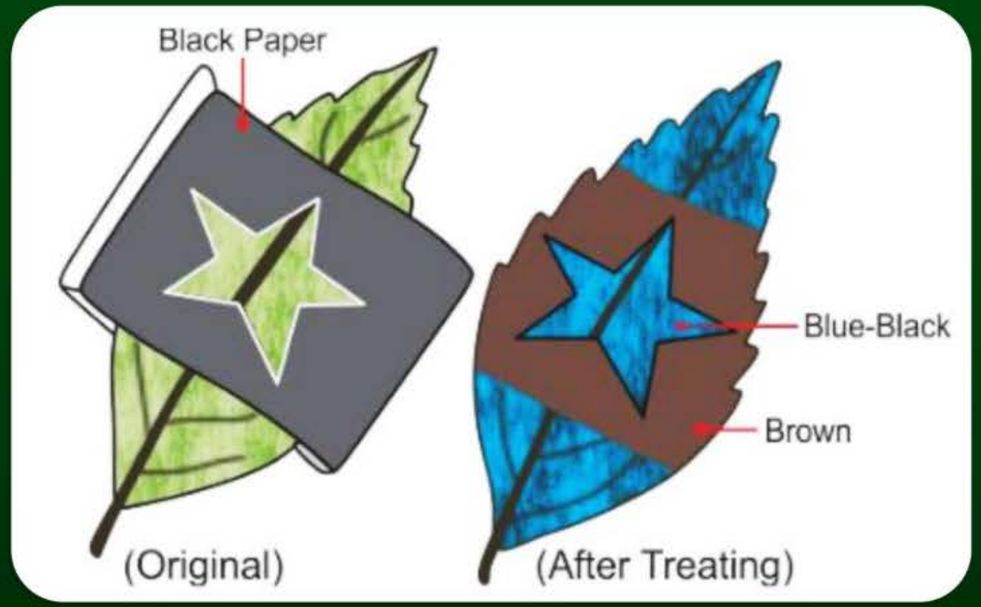
Sunlight

Green Part → Photosynthesize.

Non Green Part → X

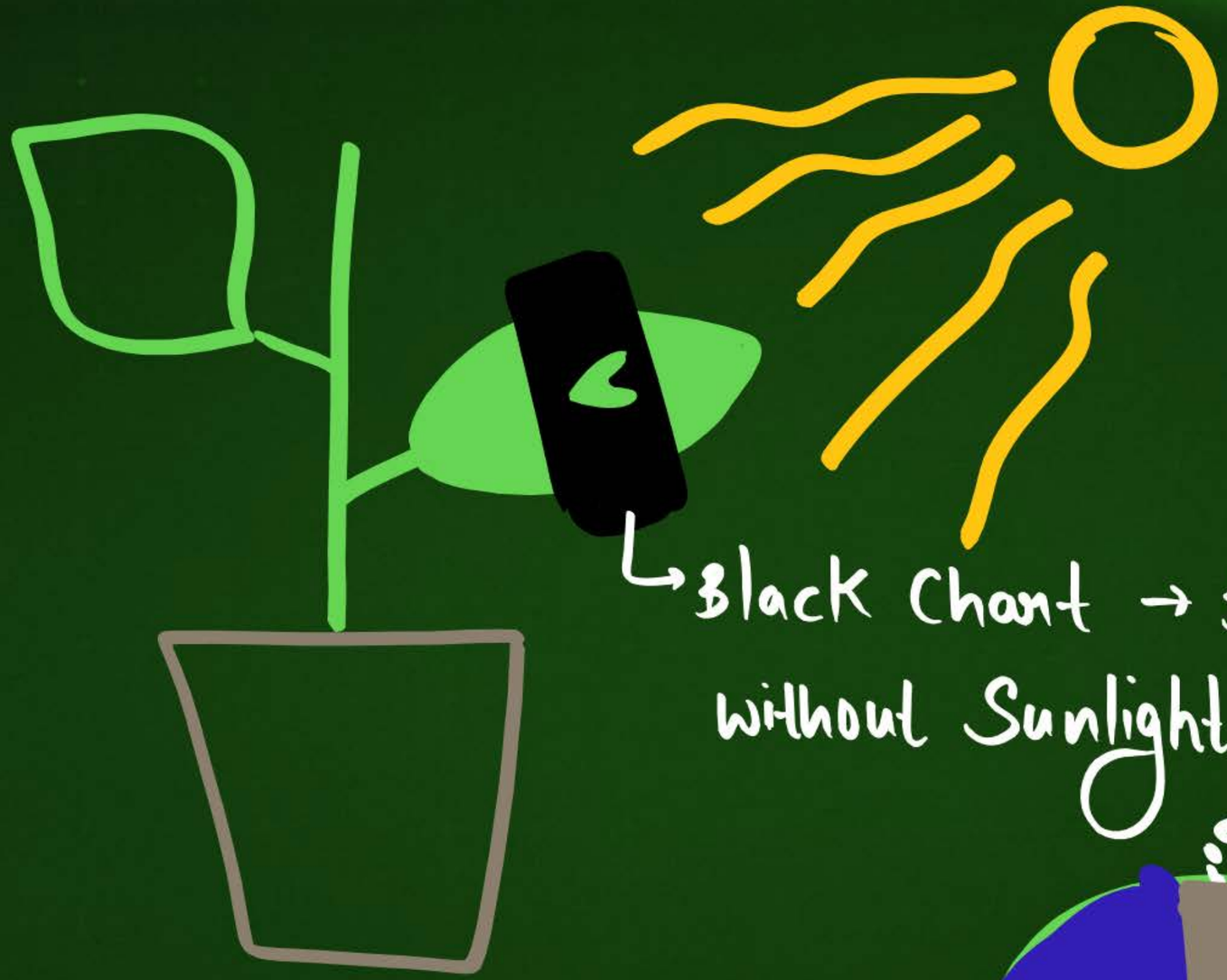
Experiment 2 : To Show that Sunlight is Necessary for Photosynthesis

- Take a plant with destarched leaves.
- Cover one of its leaves with black paper on which a design is cut.
- Place this plant in the sun.
- After a few hours, test the leaf which is covered by black paper for the presence of starch.



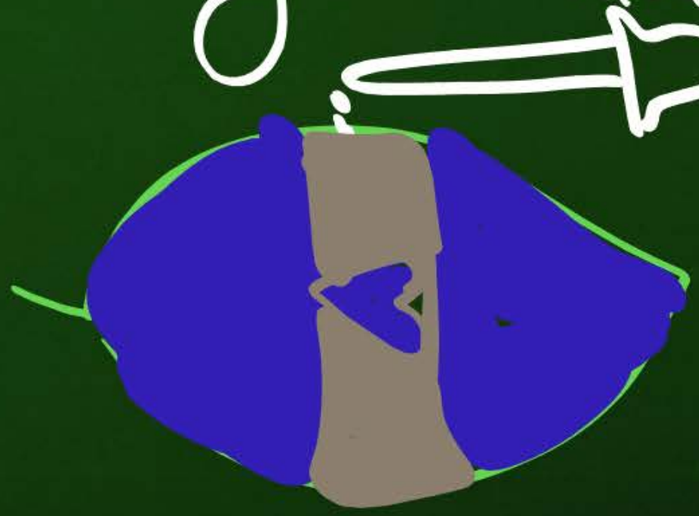
Leaf before treatment

Leaf after treatment with iodine solution



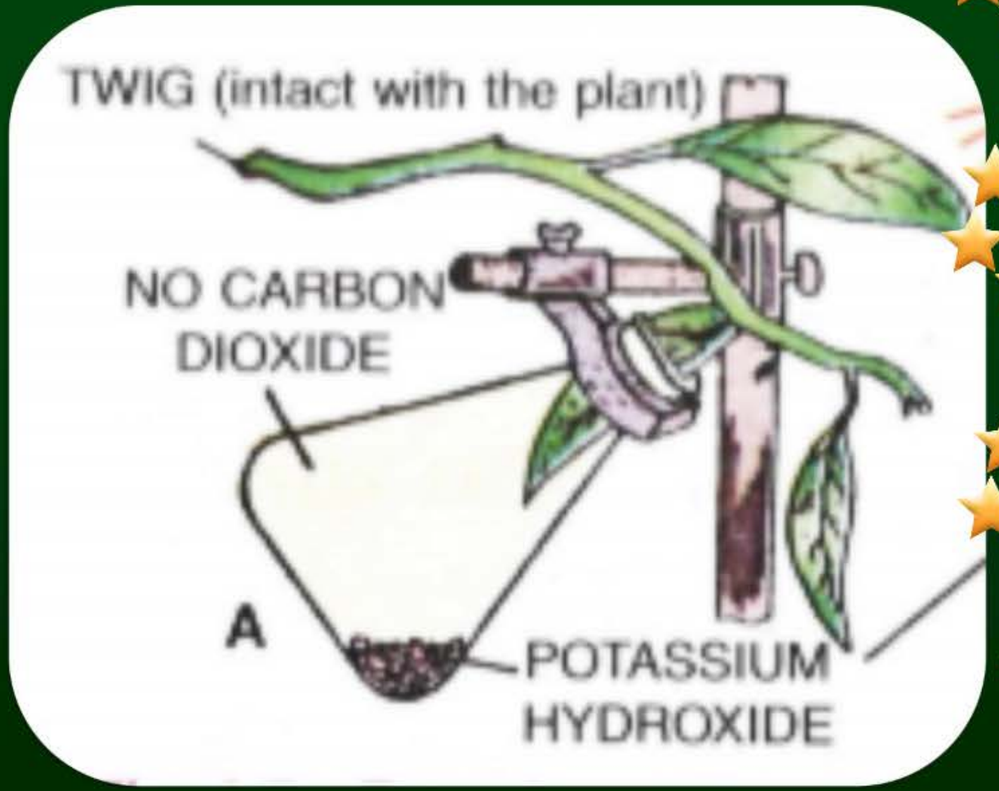
↳ Black Chart → Blocks Sunlight

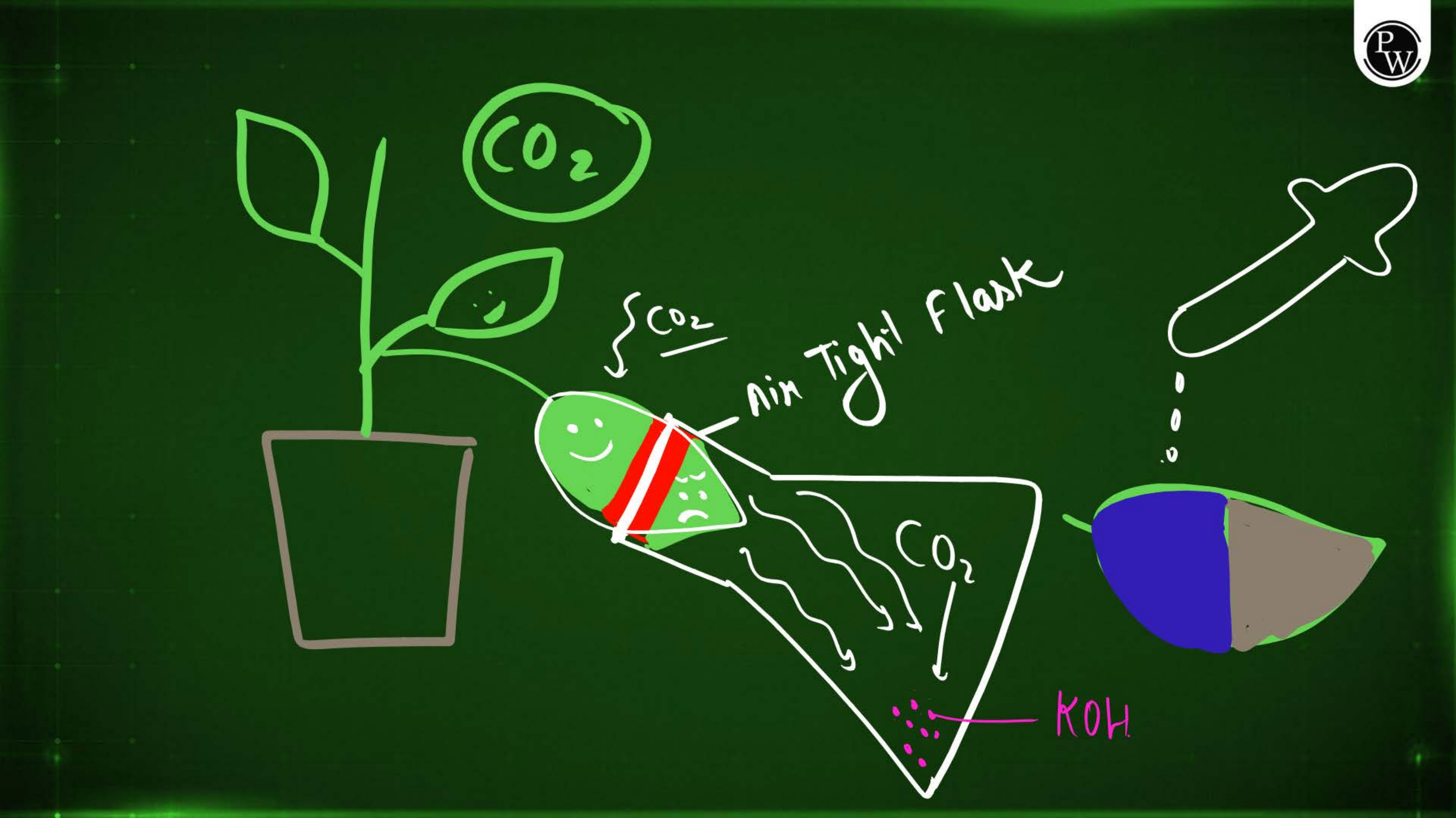
Without Sunlight No Photosynthesis



Experiment 3: To show that carbon dioxide is necessary for photosynthesis.

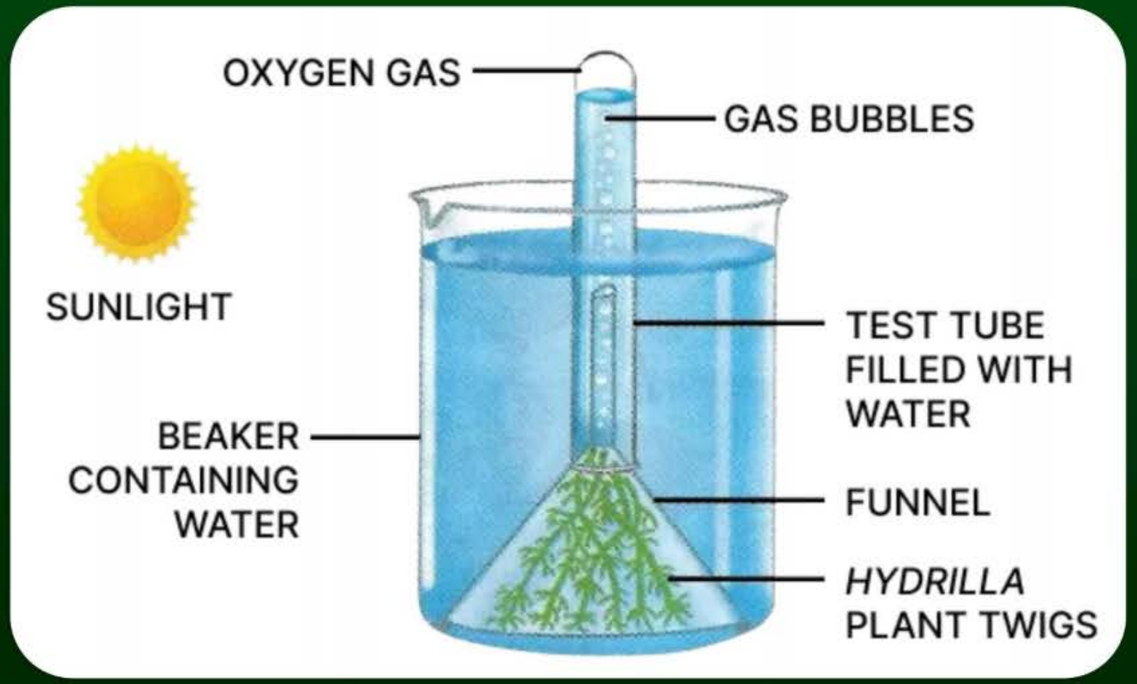
- Take a plant with destarched leaves.
- Insert a part (half) of one of its leaves (through a split cork) into a conical flask containing potassium hydroxide (Potassium hydroxide absorbs carbon dioxide).
- Leave the plant in sunlight.
- After a few hours, test the leaf for starch.



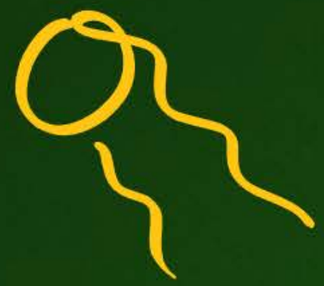

CO_2
CO_2
Air Tight Flask
CO_2
KOH

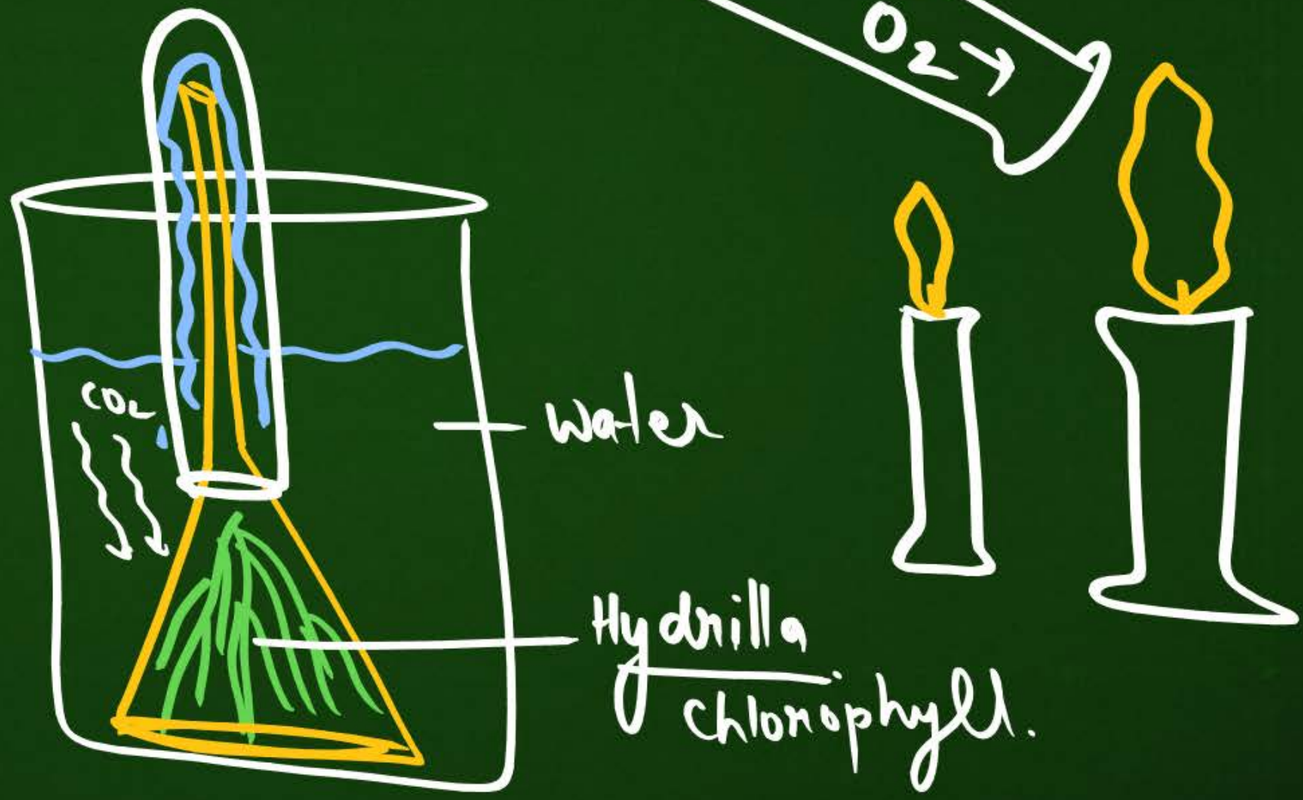
Experiment 4: To show that oxygen is produced during photosynthesis.

- Place the apparatus in the sun for a few hours. Bubbles of the gas will collect in the test-tube.
- Test the gas in the test-tube. A glowing splinter bursts into flame which shows the presence of oxygen.



$$CO_2 + H_2O \xrightarrow[\text{Sunlight}]{\text{Chlorophyll}} C_6H_{12}O_6 + H_2O + O_2$$





Carbon Cycle

The essential steps in the carbon cycle are as follows :

(i) Photosynthesis :
Green plants (producers) use carbon dioxide of the air to produce carbohydrates (sugar, starch, etc.)

$CO_2 + H_2O \xrightarrow[\text{Sun}]{\text{ch}} C_6H_{12}O_6 \ldots -$

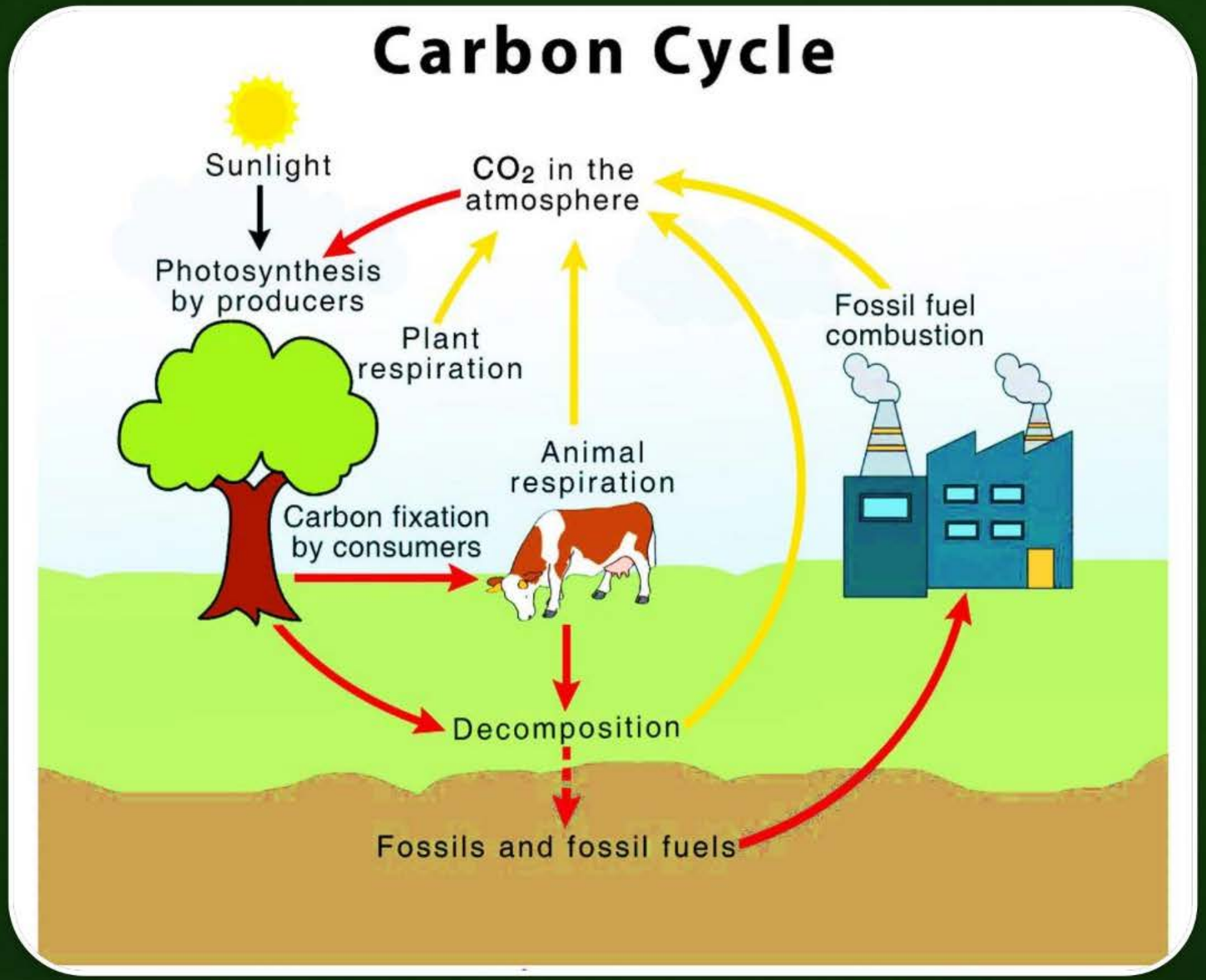
(ii) Food chains : ✓ Respiration.
Through food chains, the food (containing carbon in the carbohydrates and other products) passes on from plants to animals(consumers). Carbon is an essential element of all body tissues of both plants and animals.

green plants ⟶ Consumers [Herbivores, Carnivores]

$C_6H_{12}O_6$

$C_6H_{12}O_6 + O_2 \longrightarrow CO_2 \uparrow$

Carbon Cycle



Carbon Cycle

(iii) Respiration :

All plants and animals, respire by oxidising carbohydrates in their cells to produce energy and give out carbon dioxide into the atmosphere.

(iv) Decay:

The dead remains of plants and animals are consumed by bacteria and fungi (decomposers), and in the process, they break down the organic matter, releasing carbon dioxide back into the atmosphere.

Carbon Cycle

$CH_4 + O_2 \rightarrow CO_2$.

(v) Combustion (burning) :

When a fuel such as wood or a fossil fuel like coal, petroleum or natural gas, is burnt, the carbon contained in it is oxidised to carbon dioxide, which is given back into the atmosphere. All the fuels named above originally come from living organisms.

(vi) Heating limestone : ✓

A certain amount of carbon dioxide is released during heating or burning of limestone in lime kilns.

$CaCO_3 \longrightarrow CaO + CO_2\uparrow$

QUESTION

Photosynthesis is affected by following factors :

1. CO_2 + temperature + light
2. CO_2 + light + N_2
3. Water + temperature + N_2
4. More than one of the above



QUESTION

Which one of the following is NOT a requirement for photosynthesis?

1. Chlorophyll
2. CO_2
3. O_2 ✓
4. More than one of the above



QUESTION

Root → Starch Stored

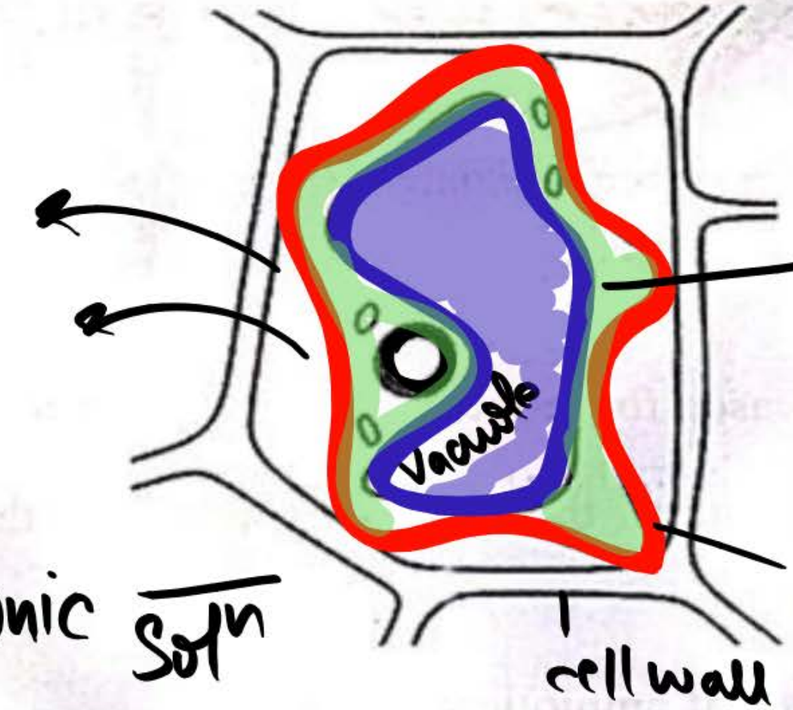
Sweet potato is an underground structure which stores food. Where is the food prepared in this plant?

(leaf)

1. Leaves
2. Fruit
3. Stem ✓
4. Root



(v) Given below is the diagram of a *plasmolysed* cortical cell of a root. [3]



2025.

cytoplasm

× P

Hypertonic Soln

cell Membrane/ plasma membrane

cell wall

Copy the diagram and label the following parts in the diagram:

(a) Cytoplasm

(b) Hypertonic solution

(c) Plasma membrane

Question 4

(i) What is the structure of DNA as proposed by Watson and Crick? [1]

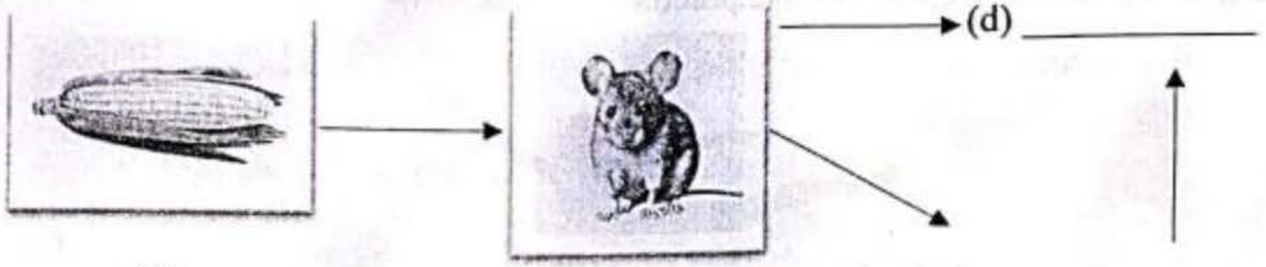
Double helical

(ii) Name the white blood cells that: [2]

(a) show phagocytosis

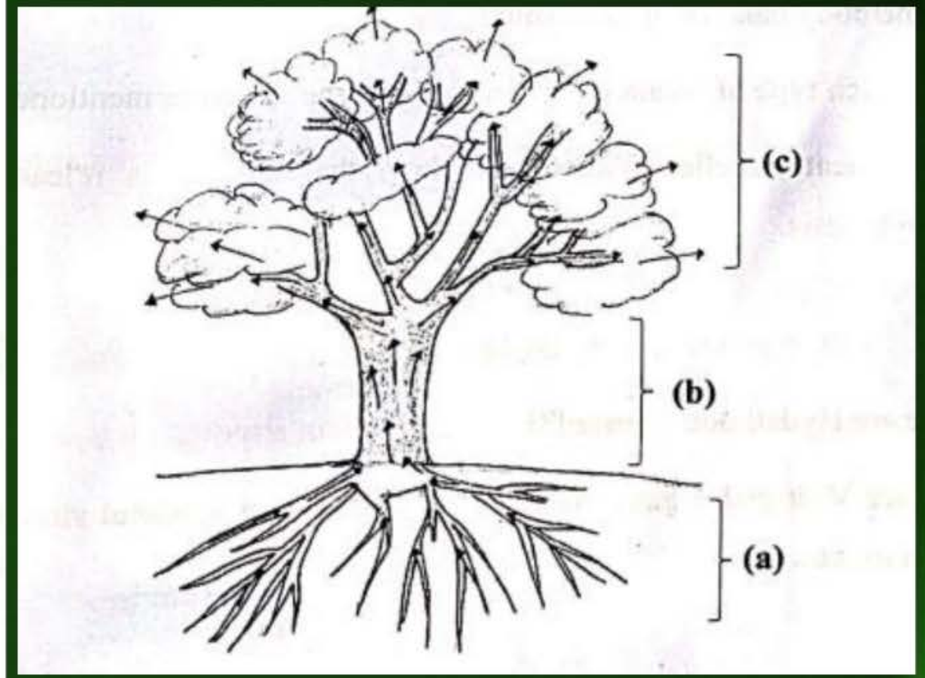
(b) produce antibodies

(iii) Complete the food chain by mentioning the names of the organisms in (c) and (d): [2]



(iv) State *any two* harmful effects of acid rain. [2]

(v) Given below is a picture of a tree. Mention the scientific phenomena occurring in the parts labelled (a), (b) and (c). [3]



(vii) The optimum temperature for photosynthesis is:

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(a) 10 – 15°C

(b) 60 – 65°C

(c) 40 – 45°C

(d) 25 – 35°C

(viii) Ajay and Vijay are siblings. Ajay can roll his tongue while Vijay cannot. What is the genotype of their parents?

(a) Both the parents are homozygous dominant.

(b) Both the parents are homozygous recessive.

(c) One parent is homozygous dominant, and the other is homozygous recessive.

(d) Both the parents are heterozygous dominant.

Stimuli and Response

- Stimuli is a change in either external or internal environment of an organism.
- Response is the resulting actions or movements caused by stimuli.



Phytohormones

- Plant sensitivity and coordination is controlled by some chemicals produced by them.
- Plants respond to stimuli by producing chemical compounds called "hormones" that work as messengers.
- The term hormone was first used by William Bayliss and Ernest Starling in 1902.



Phytohormones

- Because of some hormones plant growth is stimulated or maybe inhibited.
- The main areas of growth (cell division) in plants are meristems as they are sensitive to hormones.
- The hormone helps to stretch the cellulose wall of the meristematic cells to facilitate division.

Phytohormones

- Plants are sensitive to light and respond to the direction of light. Also sensitive to water, temperature, gravity, etc.
- Shoot grows towards light, but roots grow away from it.
- Phytohormone was coined to distinguish plant from animal hormones.

Thank You